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3D filter and fabrication method thereof

Abstract

A three-dimensional (3D) filter and a fabrication method thereof are provided. The 3D filter includes a circuit board, a first ring resonator, a second ring resonator and a via structure. The circuit board includes a first circuit layer, a second circuit layer and a third circuit layer. The third circuit layer is located between the first circuit board and the second circuit board. The first ring resonator is disposed in the first circuit layer and has a first ring surrounded area corresponding to a first cut-off frequency band. The second ring resonator is disposed in the second circuit layer and has a second ring surrounded area corresponding to a second cut-off frequency band. The via structure passes through the first circuit layer, the second circuit layer and the third circuit layer, and is electrically connected to the first ring resonator and the second ring resonator.

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priority to Taiwan Application Serial Number 111109454 filed Mar. 15, 2022, which is herein incorporated by reference.

BACKGROUND

Technical Field

(2) The invention relates to a three-dimensional (3D) filter and a fabrication method thereof.

Description of Related Art

(3) Transmission frequency of wireless signals is getting higher and higher as along with the development of wireless communication technology. Therefore, filters corresponding to higher transmission frequencies are required to meet the wireless communication technology. Especially when the filters are designed to meet requirements of electrical characteristics in a limited product area. For example, existing antenna filter in package (AFIP) is required to design the antenna filter in a small plane area. In addition, most filters designed in the millimeter wave frequency bands cannot maintain a transmission loss less than −1 dB in the passband and flatness of the filters are not considered when the filters are designed. The linearity of the amplifier in a corresponding radio transmitting system is affected when the flatness of the filter is not managed.

(4) Therefore, a filter can be easily disposed in a limited surface area with low loss and high flatness characteristics is needed.

SUMMARY

(5) Embodiments of the invention provide a three-dimensional (3D) filter and a fabrication method thereof. The filter adopts a 3D stagger structure to achieve the purpose of having a smaller plane area and taking into account low loss and high flatness characteristics.

(6) According to the present invention, the 3D filter includes a circuit board, a first ring resonator, a second ring resonator and a via structure. The circuit board includes a first circuit layer, a second circuit layer and a third circuit layer. The second circuit layer is located on the first circuit layer, and the third circuit layer is located between the first circuit layer and the second circuit layer. The first ring resonator is disposed in the first circuit layer of the circuit board, in which the first ring resonator has a first ring surrounded area and corresponds to a first cut-off frequency band. The second ring resonator is disposed in the second circuit layer of the circuit board, in which the second ring resonator has a second ring surrounded area and corresponds to a second cut-off frequency band. The via structure passes through the first circuit layer, the second circuit layer and the third circuit layer, and is electrically connected to the first ring resonator and the second ring resonator. The first ring surrounded area of the first ring resonator is different in size from the second ring surrounded area of the second ring resonator, and the first cut-off frequency band is different in frequency from the second cut-off frequency band. The first ring resonator corresponds to one of a higher cut-off frequency band and a lower cut-off frequency band, and the second ring resonator corresponds to the other one of the higher cut-off frequency band and the lower cut-off frequency band.

(7) In some embodiments, the 3D filter further includes: a first signal feed-in line, a first signal output line, a second signal feed-in line and a second signal output line. The first signal feed-in line is disposed in the second circuit layer of the circuit board, and is electrically connected to the second ring resonator. The first signal output line is disposed in the second circuit layer of the circuit board, and is electrically connected between the second ring resonator and the via structure. The second signal feed-in line is disposed in the first circuit layer of the circuit board, and is electrically connected between the first ring resonator and the via structure. The second signal output line is disposed in the first circuit layer of the circuit board, and is electrically connected to the first ring resonator. Extension directions of the first signal feed-in line and the second signal feed-in line are perpendicular to each other, and extension directions of the first signal output line and the second signal output line are perpendicular to each other to form an orthogonal feed-in structure. The first signal feed-in line is used as one of an input port and an output port of the 3D filter, and the second signal output line is used as another one of the input port and the output port of the 3D filter.

(8) In some embodiments, the 3D filter further includes: a first open stub, a second open stub, a third open stub and a fourth open stub. The first open stub is disposed in the first circuit layer of the circuit board and is electrically connected to the first ring resonator. The second open stub is disposed in the first circuit layer of the circuit board and is electrically connected to the first ring resonator. The third open stub is disposed in the second circuit layer of the circuit board and is electrically connected to the second ring resonator. The fourth open stub is disposed in the second circuit layer of the circuit board and is electrically connected to the second ring resonator.

(9) In some embodiments, the first ring resonator has a first side, a second side, a third side and a fourth side. The first side is opposite to the third side, and the second side is opposite to the fourth side. The second signal output line is adjacent to the second side. The second signal feed-in line is adjacent to the first side. The first open stub is adjacent to the fourth side. The second open stub is adjacent to the third side.

(10) In some embodiments, the second ring resonator has a fifth side, a sixth side, a seventh side and an eighth side. The fifth side is opposite to the seventh side, and the sixth side is opposite to the

eighth side. The first signal feed-in line is adjacent to the eighth side. The first signal output line is adjacent to the fifth side. The third open stub is adjacent to the seventh side. The fourth open stub is adjacent to the sixth side.
(11) In some embodiments, the first open stub, the second open stub, the third open stub and the fourth open stub are quarter-wavelength open stubs.
(12) In some embodiments, the first signal feed-in line has a first end portion, a second end portion and a middle portion, the first end portion is adjacent to the second ring resonator, the second end portion is distal from the second ring resonator, the middle portion is located between the first end portion and the second end portion, and a width of the first signal output line gradually widens from the first end portion to the second end portion to provide a stepped impedance.
(13) In some embodiments, the 3D filter further includes: a first metal patch and a second metal patch. The first metal patch is disposed in the first circuit layer of the circuit board and is located in the first ring surrounded area of the first ring resonator. The second metal patch is disposed in the second circuit layer of the circuit board and is located in the second ring surrounded area of the second ring resonator.
(14) In some embodiments, the 3D filter further includes: an insulating material disposed around the opening to prevent the via structure from being electrically connected to the ground metal layer.
(15) In some embodiments, the 3D filter further includes a ground metal layer. The ground metal layer is disposed in the third circuit layer as a common reference ground plane of the first ring resonator and the second ring resonator. The ground metal layer has an opening, and the via structure passes through the opening to vertically penetrate the ground metal layer.
(16) In some embodiments, the circuit board further includes a first liquid crystal polymer (LCP) layer and a second liquid crystal polymer layer, wherein the first LCP layer is located between the first circuit layer and the second circuit layer, the second LCP layer is located between the second circuit layer and the third circuit layer, and the via structure further penetrates the first LCP layer and the second LCP layer.
(17) According to an embodiment of the present invention, the fabrication method of the 3D filter includes: forming a first ring resonator in the first circuit layer according to a predetermined frequency band of the 3D filter, and forming a second ring resonator in the second circuit layer, in which the first ring resonator has a first ring surrounded area corresponding to a first cut-off frequency band of the predetermined frequency band, the second ring resonator has a second ring surrounded area corresponding to a second cut-off frequency band of the predetermined frequency band, and the first ring surrounded area is different in size from the second ring surrounded area; forming a ground metal layer in the third circuit layer, in which the third circuit layer is located between the first circuit layer and the second circuit layer, and the ground metal layer serves as a common reference ground plane for the first ring resonator and the second ring resonator; and forming a via structure in the first circuit layer, the second circuit layer and the third circuit layer, in which the via structure passes through the first circuit layer, the second circuit layer and the third circuit layer to electrically connected to the first ring resonator and the second ring resonator.
(18) In some embodiments, the method for fabricating the 3D filter further includes: forming a first signal feed-in line, a first signal output line, a third open stub, a fourth open stub and a second metal patch in the second circuit layer according to the predetermined frequency band of the 3D filter.
(19) In some embodiments, the method for fabricating the 3D filter further includes: forming a second signal feed-in line, a second signal output line, a first open stub, a second open stub and a first metal patch in the first circuit layer according to the predetermined frequency band of the 3D filter.
(20) In some embodiments, the method for fabricating the 3D filter further includes: forming an opening in the ground metal layer, wherein the via structure passes through the opening to vertically penetrate the ground metal layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic diagram showing the structure of a three-dimensional (3D) filter according to an embodiment of the present invention.

(2) FIG. **2** is a schematic diagram showing the structure of circuit board of a 3D filter.

(3) FIG. **3** is a schematic diagram showing the structure of a second ring filter according to an embodiment of the present invention.

(4) FIG. **4** is a schematic diagram showing the structure of a first ring filter according to an embodiment of the present invention.

(5) FIG. **5** is a schematic diagram showing the structure of a second ring filter according to another embodiment of the present invention.

(6) FIG. **6** is a schematic diagram showing the structure of a second ring filter according to another embodiment of the present invention.

(7) FIG. **7** is a schematic diagram showing the structure of a second ring filter according to another embodiment of the present invention.

(8) FIG. **8** is a schematic diagram showing the structure of a second ring filter according to another embodiment of the present invention.

(9) FIG. **9** is a schematic diagram showing the structure of an antenna module according to an embodiment of the present invention.

(10) FIG. **10** is a schematic flowchart showing a method for manufacturing a 3D filter according to an embodiment of the present invention.

DETAILED DESCRIPTION

(11) Terms “first”, “second”, etc. used herein do not specifically refer to particular order or sequence, but to distinguish elements or operations described with the same technical terms.

(12) Referring to FIG. **1**, a schematic structural diagram showing a three-dimensional (3D) filter **100** in accordance with an embodiment of the present invention. The 3D filter **100** includes a first ring filter **110**, a second ring filter **120**, and a via structure **130** electrically connecting the first ring filter **110** and the second ring filter **120**, in which a ground metal layer **140** is disposed between the first ring filter **110** and the second ring filter **120**. The first ring filter **110** and the second ring filter **120** are disposed on opposite sides of the ground metal layer **140**, the via structure **130** penetrates the ground metal layer **140** to electrically connect the first ring filter **110** and the second ring filter **120**, in which the via structure **130** is not electrically connected to the ground metal layer **140**.

(13) Referring to FIG. **2**, a schematic structural diagram showing a circuit board **200** of the 3D filter **100** in accordance with an embodiment of the present invention. The circuit board **200** includes a first circuit layer **210**, a second circuit layer **220**, a third circuit layer **230**, a first insulating layer **240** and a second insulating layer **250**. The second circuit layer **220** and the third circuit layer **230** are located on the first circuit layer **210**, and the third circuit layer **230** is located between the first circuit layer **210** and the second circuit layer **220**. The first insulating layer **240** and the second insulating layer **250** are disposed on the first circuit layer **210**, in which the first insulating layer **240** is disposed between the first circuit layer **210** and the third circuit layer **230** to provide electrical insulation between the first circuit layer **210** and the third circuit layer **230**. The second insulating layer **250** is disposed between the third circuit layer **230** and the second circuit layer **220** to provide electrical insulation between the third circuit layer **230** and the second circuit layer **220**.

(14) In this embodiment, the first circuit layer **210**, the second circuit layer **220** and the third circuit layer **230** are made of copper foil, and the first insulating layer **240** and the second insulating layer **250** are made of liquid crystal polymer (LCP) layers, but other embodiments of the present invention are not limited thereto. Other conductive materials and insulating materials may be used

to form the first circuit layer **210**, the second circuit layer **220**, the third circuit layer **230**, the first insulating layer **240** and the second insulating layer **250** according to actual demands. For example, the first circuit layer **210**, the second circuit layer **220** and the third circuit layer **230** may be made of silver, gold, aluminum, nickel, iron or a compound of the above materials, and the first insulating layer **240** and the second insulating layer **250** may be made of polyimide (PI) or modified polyimide (MPI).

(15) Referring to FIG. **1** and FIG. **2**, the first ring filter **110** is located in the first circuit layer **210**; the second ring filter **120** is located in the second circuit layer **220**; the ground metal layer **140** is located in the third circuit layer **230**; the via structure **130** passes through the first circuit layer **210**, the second circuit layer **220** and the third circuit layer **230** in this embodiment. The via structure **130** may be made by using vias or blind holes in some embodiments.

(16) Referring to FIG. **3**, a schematic structural diagram showing the second ring filter **120** according to an embodiment of the invention, in which the second ring filter **120** is located in the second circuit layer **220**. The second ring filter **120** includes a ring resonator **310**, a first signal feed-in line **320** and a first signal output line **330**. The ring resonator **310** has a ring surrounded area A**1** with sides electrically connected to the first signal feed-in line **320** and the first signal output line **330**. The first signal feed-in line **320** is used as an input port of the 3D filter **100** to receive a high-frequency input signal to be processed. The first signal output line **330** is electrically connected between the ring resonator **310** and an end portion **131** of the via structure **130** to provide an output signal to the via structure **130**.

(17) Referring to FIG. **4**, a schematic structural diagram showing the first ring filter **110** according to an embodiment of the invention, in which the first ring filter **110** is located in the first circuit layer **210**. The first ring filter **110** includes a ring resonator **410**, a second signal feed-in line **420** and a second signal output line **430**. The ring resonator **410** has a ring surrounded area A**2**, and in this embodiment, the ring surrounded area A**2** is smaller than the ring surrounded area A**1**. In other words, the first ring filter **110** is a small ring filter and the second ring filter **120** is a large ring filter. The second signal feed-in line **420** is electrically connected between the ring resonator **410** and the other end portion **132** of the via structure **130**, and receives the output signal provided by the second ring filter **120** through the via structure **130**. The second signal output line **430** is electrically connected to the ring resonator **410** to serve as an output port of the 3D filter **100**. Therefore, the first ring filter **110** can output a high-frequency filter signal from the second signal output line **430** corresponding to the output signal provided by the second ring filter **120**. However, the invention is not limited thereto. The first ring filter **110** may be designed to be larger than the second ring filter **120**, and the sizes of the ring areas of the two filters must be different.

(18) In addition, the 3D filter **100** provides an orthogonal feed-in structure. Specifically, extension directions of the first signal feed-in line **320** and the second signal feed-in line **420** are perpendicular to each other, and extension directions of the first signal output line **330** and the second signal output line **430** are perpendicular to each other to form the orthogonal feed-in structure. In this embodiment, the first signal feed-in line **320** is an input port of the 3D filter, and the second signal output line **430** is an output port of the 3D filter. In other embodiments, the configurations of the input port and the output port can also be reversed, so that the first signal feed-in line **320** becomes the output port, and the second signal output line **430** becomes the input port. In other words, a signal to be filtered can be input from the first signal feed-in line **320**, and then the filtered signal can be output through the second signal output line **430**. Alternatively, the signal to be filtered can be input from the second signal output line **430**, and then the filtered signal can be output through the first signal feed-in line **320**.

(19) In this embodiment, the first ring filter **110** further includes a first open stub **440**, a second open stub **450** and a first metal patch **460**. The first open stub **440** is disposed on one side of the ring resonator **410** and is opposite to one of the second signal feed-in line **420** and the second signal output line **430**, and the second open stub **450** is disposed on the other side of the ring

resonator **410** and is opposite to the other one of the second signal feed-in line **420** and the second signal output line **430**. Specifically, the ring resonator **410** has a first side **411**, a second side **412**, a third side **413** and a fourth side **414**, in which the first side **411** is opposite to the third side **413**, and the second side **412** is opposite to the fourth side **414**. The second signal output line **430** is adjacent to the second side **412**, and the first open stub **440** is adjacent to the fourth side **414**. The second signal feed-in line **420** is adjacent to the first side **411**, and the second open stub **450** is adjacent to the third side **413**. The first metal patch **460** is disposed in the ring surrounded area A**2** and is disposed corresponding to the first open stub **440** and the second open stub **450**. Specifically, the first open stub **440** and the second open stub **450** are adjacent to the fourth side **414** and the third side **413**, and the first metal patch **460** is disposed in a corner corresponding to the third side **413** and the fourth side **414**. According to the above design, the first open stub **440** and the second open stub **450** can be combined with the ring resonator **410** to provide a bandpass filter, and the first metal patch **460** can provide impedance matching for the impedance discontinuity problem caused by the via structure **130**. A bandwidth of the first ring filter **110** may be affected by the size of the first metal patch **460**. The larger the size of the first metal patch **460** is, the larger the bandwidth of the first ring filter **110** is.

(20) Similarly, retuning to FIG. **3**, the second ring filter **120** further includes a third open stub **350**, a fourth open stub **340** and a second metal patch **360**. The third open stub **350** is disposed on one side of the ring resonator **310** and is opposite to one of the first signal feed-in line **320** and the first signal output line **330**, and the fourth open stub **340** is disposed on the other side of the ring resonator **310** and is opposite to the other of the first signal feed-in line **320** and the first signal output line **330**. Specifically, the ring resonator **310** has a fifth side **315**, a sixth side **316**, a seventh side **317** and an eighth side **318**, in which the fifth side **315** is opposite to the seventh side **317**, and the sixth side **316** is opposite to the eighth side **318**. The first signal feed-in line **320** is adjacent to the eighth side **318**, and the third open stub **350** is adjacent to the seventh side **317**. The first signal output line **330** is adjacent to the fifth side **315**, and the fourth open stub **340** is adjacent to the sixth side **316**. The second metal patch **360** is disposed in the ring surrounded area A**1** and is disposed corresponding to the third open stub **350** and the fourth open stub **340**. Specifically, the third open stub **350** and the fourth open stub **340** are adjacent to the sixth side **316** and the seventh side **317**, and the second metal patch **360** is disposed in a corner corresponding to the sixth side **316** and the seventh side **317**. According to the above design, the third open stub **350** and the fourth open stub **340** can be combined with the ring resonator **310** to provide a bandpass filter, and the second metal patch **360** can provide impedance matching for the impedance discontinuity problem caused by the via structure **130**. A bandwidth of the second ring filter **120** can be affected by the size of the second metal patch **360**. The larger the size of the second metal patch **360** is, the larger the bandwidth of the second ring filter **120** is.

(21) In an embodiment of the invention, the wrapping length (or inner ring length) of the ring resonators **310** and **410** is n times a wavelength, where n is an integer greater than or equal to 1. The lengths of the first open stub **440**, the second open stub **450**, the third open stub **350** and the fourth open stub **340** are quarter-wavelengths. In addition, the ring resonators **310** and **410** are octagonal in the embodiment, but the embodiments of the invention are not limited thereto. In other embodiments of the present invention, the ring resonators **310** and **410** have shapes of up-down symmetry and bilateral symmetry, such as rectangular or hexagonal.

(22) From the above description, it may be understood that the 3D filter **100** in embodiments of the present invention utilize the superposition of two passbands provided by the large ring filter (i.e. the second ring filter **120**) and the small ring filter (i.e. the first ring filter **110**) to provide a composite passband, in which a higher cut-off frequency of the composite passband of the 3D filter **100** is determined by the higher frequency passband provided by the second ring filter **120**, and a lower cut-off frequency of the composite passband is determined by the lower frequency passband provided by the first ring filter **110**. In this embodiment, the higher cut-off frequency of the

composite passband is 64 GHZ, and the lower cut-off frequency of the composite passband is 54 GHZ, so an operating frequency is 60 GHz. However, embodiments of the present invention are not limited thereto. The first ring filter **110** and second ring filter **120** may be adjusted, so that the composite passband of the 3D filter **100** may meet different demands. For example, in embodiments of the present invention, the large ring filter (i.e. the second ring filter **120**) may correspond to one of the higher cut-off frequency and the lower cut-off frequency (such as the higher cut-off frequency), and the small ring filter (i.e. the first ring filter **110**) may correspond to the other one of the higher cut-off frequency and the lower cut-off frequency (such as the lower cut-off frequency). In some embodiments, the operating frequency of the 3D filter **100** may be 56.13-62.61 GHz.

(23) Furthermore, the first ring filter **110** and the second ring filter **120** are made into a 3D connection structure in the 3D filter **100** of the embodiment of the present invention, in which the first ring filter **110** and the second ring filter **120** are electrically connected by the via structure **130** penetrating through the ground metal layer **140**. Since the first ring filter **110** and the second ring filter **120** adopt the 3D connection structure, the 3D filter **100** has a smaller plane area.

(24) Referring to FIG. **5**, a schematic structural diagram showing a second ring filter **120** according to another embodiment of the present invention. In this embodiment, the shape of the ring resonator **310** of the second ring filter **120** is changed, and it further has a plurality of curved sides BS**1**-BS**4**, in which the curved side BS**1** is located between the fifth side **315** and the eighth side **318**; the curved side BS**2** is located between the seventh side **317** and the eighth side **318**; the curved side BS**3** is located between the sixth side **316** and the seventh side **317**; the curved side BS**4** is located between the fifth side **315** and the sixth side **316**. Through such a design, the length of the ring resonator **310** can be increased in a limited plane area, thereby changing the size and passband range provided by the 3D filter **100**.

(25) Similarly, the length of the ring resonator **410** of the first ring filter **110** can also be changed through the settings of curved sides. Since the details of the setting have been described in the above paragraphs, they will not be repeated here.

(26) Referring to FIG. **6**, a schematic structural diagram showing a second ring filter **120** according to another embodiment of the present invention. In this embodiment, the shapes of the third open stub **350** and the fourth open stub **340** of the second ring filter **120** are modified to further have bent portions **340**B and **350**B. Through such a design, the lengths of the third open stub **350** and the fourth open stub **340** can be increased in a limited plane area, thereby changing the size and passband range provided by the 3D filter **100**. Similarly, the lengths of the first open stub **440** and the second open stub **450** of the first ring filter **110** can also be changed through the settings of the bent portions.

(27) Referring to FIG. **7**, is a schematic structural diagram showing a second ring filter **120** according to another embodiment of the present invention. In this embodiment, the shape of the first signal feed-in line **320** of the second ring filter **120** is changed so that it has a stepped width change. Specifically, the first signal feed-in line **320** includes a first end portion **320***a*, a second end portion **320***b* and a middle portion **320***c*. The first end portion **320***a* is adjacent to the ring resonator **310**, the second end portion **320***b* is distal from the ring resonator **310**, the middle portion **320***c* is located between the first end portion **320***a* and the second end portion **320***b*, and the width of the first signal feed-in line **320** gradually widens from the first end portion **320***a* to the second end portion **320***b* to provide a stepped impedance.

(28) Referring to FIG. **8**, a schematic structural diagram showing a second ring filter **120** according to another embodiment of the present invention. In this embodiment, the shape of the first signal feed-in line **320** of the second ring filter **120** is changed so that it has a gradual width change. Specifically, the widths of the first end portion **320***a*, the second end portion **320***b*, and the middle portion **320***c* of the first signal feed-in line **320** change gradually, instead of the stepped change as shown in FIG. **7**.

(29) Referring to FIG. **9**, a schematic structural diagram showing an antenna module **900** according to an embodiment of the present invention. The antenna module **900** includes a radio frequency chip (RF IC) **910**, an antenna device **920**, an antenna device **930** and a circuit board **940**. The radio frequency chip **910**, the antenna device **920** and the antenna device **930** are disposed on the circuit board **940**, in which the circuit board **940** may have a multi-layer circuit structure to form various circuits. For example, the structure of the circuit board **200** is included to provide the 3D filter **100** (Referring to FIG. **1** and FIG. **2** together). In the embodiment, the antenna module **900** can be an antenna filter in package (AFIP). Compared with the conventional system in package (SiP), the antenna package of this embodiment is further integrated with the antenna device and the filter and has a smaller volume. In some embodiments, the circuits of the circuit board **200** may be used to form the antenna device, so that the antenna device **920** and the antenna device **930** can be omitted, and the space occupied by the antenna module **900** may be further reduced.
(30) Referring to FIG. **10**, a schematic flowchart showing a fabrication method **1000** of the 3D filter **100** according to an embodiment of the present invention. The fabrication method **1000** of the 3D filter **100** can be performed using, for example, automatic antenna production equipment. The antenna automatic production equipment may include a memory and a processor, in which the memory stores instructions, and the processor can load the instructions to perform the fabrication method **1000**.
(31) In the fabrication method **1000**, step **1100** is first performed to form the ring resonator **410** in the first circuit layer **210** and to form the ring resonator **310** in the second circuit layer **220** according to a predetermined frequency band required by the 3D filter **100**, in which the ring resonator **410** has a ring surrounded area A**2** corresponding to the first cut-off frequency band of the predetermined frequency band (corresponding to the lower cut-off frequency), the ring resonator **310** has a ring surrounded area A**1** corresponding to the second cut-off frequency band of the predetermined frequency band (corresponding to a higher cut-off frequency), and the ring surrounded area A**2** is smaller than the ring surrounded area A**1**.
(32) Then, step **1200** is performed to form the first signal feed-in line **320**, the first signal output line **330**, the third open stub **350**, the fourth open stub **340** and the second metal patch **360** in the second circuit layer **220** according to the predetermined frequency band of the 3D filter **100**.
(33) Then, step **1300** is performed to form the second signal feed-in line **420**, the second signal output line **430**, the first open stub **440**, the second open stub **450** and the first metal patch **460** in the first circuit layer **210** according to the predetermined frequency band of the 3D filter **100**.
(34) Then, step **1400** is performed to form a ground metal layer **140** in the third circuit layer **230** to serve as a common reference ground plane of the ring resonators **310** and **410**.
(35) Next, step **1500** is performed to form the via structure **130** in the first circuit layer **210**, the second circuit layer **220** and the third circuit layer **230**, in which the via structure **130** passes through the first circuit layer **210**, the second circuit layer **220** and the third circuit layer **230** to electrically connect the ring resonators **310** and **410**.
(36) In some embodiments, the fabrication method **1000** forms an opening in the ground metal layer **140** so that the via structure **130** passes through the opening to electrically connect the ring resonators **310** and **410**. In addition, an insulating material is disposed around the opening to prevent the via structure **130** from being electrically connected to the ground metal layer **140**.
(37) Although the present invention has been disclosed above with embodiments, it is not intended to limit the present invention. Those skilled in the art may make some changes and modifications without departing from the spirit and scope of the present invention. Therefore, the scope of protection of the present invention should be defined by the scope of the appended patent application.

Claims

1. A three-dimensional (3D) filter, comprising: a circuit board comprising a first circuit layer, a second circuit layer and a third circuit layer, wherein the second circuit layer is located on the first circuit layer, and the third circuit layer is located between the first circuit layer and the second circuit layer; a first ring resonator disposed in the first circuit layer of the circuit board, wherein the first ring resonator has a first ring surrounded area corresponding to a first cut-off frequency band; a second ring resonator disposed in the second circuit layer of the circuit board, wherein the second ring resonator has a second ring surrounded area corresponding to a second cut-off frequency band; and a via structure passing through the first circuit layer, the second circuit layer and the third circuit layer, and electrically connected to the first ring resonator and the second ring resonator; wherein the first ring surrounded area of the first ring resonator is different in size from the second ring surrounded area of the second ring resonator, and the first cut-off frequency band is different in frequency from the second cut-off frequency band; the first ring resonator corresponds to one of a higher cut-off frequency band and a lower cut-off frequency band, and the second ring resonator corresponds to the other one of the higher cut-off frequency band and the lower cut-off frequency band.
2. The 3D filter of claim 1, further comprising: a first signal feed-in line disposed in the second circuit layer of the circuit board, and electrically connected to the second ring resonator; a first signal output line disposed in the second circuit layer of the circuit board, and electrically connected between the second ring resonator and the via structure; a second signal feed-in line disposed in the first circuit layer of the circuit board, and electrically connected between the first ring resonator and the via structure; and a second signal output line disposed in the first circuit layer of the circuit board, and electrically connected to the first ring resonator; wherein extension directions of the first signal feed-in line and the second signal feed-in line are perpendicular to each other, and extension directions of the first signal output line and the second signal output line are perpendicular to each other to form an orthogonal feed-in structure; wherein the first signal feed-in line is used as one of an input port and an output port of the 3D filter, and the second signal output line is used as another one of the input port and the output port of the 3D filter.
3. The 3D filter of claim 2, further comprising: a first open stub disposed in the first circuit layer of the circuit board and electrically connected to the first ring resonator; a second open stub disposed in the first circuit layer of the circuit board and electrically connected to the first ring resonator; a third open stub disposed in the second circuit layer of the circuit board and electrically connected to the second ring resonator; and a fourth open stub disposed in the second circuit layer of the circuit board and electrically connected to the second ring resonator.
4. The 3D filter of claim 3, wherein: the first ring resonator has a first side, a second side, a third side and a fourth side, the first side is opposite to the third side, and the second side is opposite to the fourth side; the second signal output line is adjacent to the second side; the second signal feed-in line is adjacent to the first side; the first open stub is adjacent to the fourth side; and the second open stub is adjacent to the third side.
5. The 3D filter of claim 3, wherein: the second ring resonator has a fifth side, a sixth side, a seventh side and an eighth side, the fifth side is opposite to the seventh side, and the sixth side is opposite to the eighth side; the first signal output line is adjacent to the fifth side; the first signal feed-in line is adjacent to the eighth side; the third open stub is adjacent to the seventh side; and the fourth open stub is adjacent to the sixth side.
6. The 3D filter of claim 3, wherein the first open stub, the second open stub, the third open stub and the fourth open stub are quarter-wavelength open stubs.
7. The 3D filter of claim 2, wherein the first signal feed-in line has a first end portion, a second end portion and a middle portion, the first end portion is adjacent to the second ring resonator, the second end portion is distal from the second ring resonator, the middle portion is located between the first end portion and the second end portion, and a width of the first signal output line gradually

widens from the first end portion to the second end portion to provide a stepped impedance.

8. The 3D filter of claim 1, further comprising: a first metal patch disposed in the first circuit layer of the circuit board and located in the first ring surrounded area of the first ring resonator; and a second metal patch disposed in the second circuit layer of the circuit board and located in the second ring surrounded area of the second ring resonator.

9. The 3D filter of claim 1, further comprising: a ground metal layer disposed in the third circuit layer as a common reference ground plane of the first ring resonator and the second ring resonator; wherein the ground metal layer comprising an opening, and the via structure passes through the opening to vertically penetrate the ground metal layer.

10. The 3D filter of claim 9, further comprising: an insulating material disposed around the opening to prevent the via structure from being electrically connected to the ground metal layer.

11. The 3D filter of claim 1, wherein the circuit board further comprises a first liquid crystal polymer (LCP) layer and a second liquid crystal polymer layer, the first liquid crystal polymer layer is located between the first circuit layer and the second circuit layer, the second liquid crystal polymer layer is located between the second circuit layer and the third circuit layer, and the via structure further penetrates the first liquid crystal polymer layer and the second liquid crystal polymer layer.

12. A method for fabricating a 3D filter, comprising: forming a first ring resonator in a first circuit layer according to a predetermined frequency band of the 3D filter, and forming a second ring resonator in a second circuit layer, wherein the first ring resonator has a first ring surrounded area corresponding to a first cut-off frequency band of the predetermined frequency band, the second ring resonator has a second ring surrounded area corresponding to a second cut-off frequency band of the predetermined frequency band, and the first ring surrounded area is different in size from the second ring surrounded area; forming a ground metal layer in a third circuit layer, wherein the third circuit layer is located between the first circuit layer and the second circuit layer, the ground metal layer serves as a common reference ground plane for the first ring resonator and the second ring resonator; and forming a via structure in the first circuit layer, the second circuit layer and the third circuit layer, wherein the via structure is disposed through the first circuit layer, the second circuit layer and the third circuit layer to be electrically connected to the first ring resonator and the second ring resonator.

13. The method for fabricating the 3D filter of claim 12, further comprising: forming a first signal feed-in line, a first signal output line, a third open stub, a fourth open stub and a second metal patch in the second circuit layer according to the predetermined frequency band of the 3D filter.

14. The method for fabricating the 3D filter of claim 12, further comprising: forming a second signal feed-in line, a second signal output line, a first open stub, a second open stub and a first metal patch in the first circuit layer according to the predetermined frequency band of the 3D filter.

15. The method for fabricating the 3D filter of claim 12, further comprising: forming an opening in the ground metal layer, wherein the via structure passes through the opening to vertically penetrate the ground metal layer.